



CUBAN NATIONAL CHEMISTRY CONTEST

“The future of our country must necessarily be a future of men of sciences, must be a future of men of thinking, because that is precisely what we are creating...”

Fidel Castro

Name(s): _____ Last name(s): _____

School: _____

Municipality: _____ Province: _____

CODE: _____ Test: **DAY 2. ELEVENTH GRADE EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 11th grade. The exam has 3 problems (problem 8, 6 items; problem 9, 8 items; problem 10, 8 items) and a total of 6 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Cero in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Water ionic product at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																							
1 1A												13 3A		14 4A		15 5A		16 6A		17 7A		18 8A	
1 H 1.008	2 He 4.003																						
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18						
11 Na 22.99	12 Mg 24.31	3 B 10.81	4 C 12.01	5 N 14.01	6 O 16.00	7 F 19.00	8 Ne 20.18	9 Ar 39.95															
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80						
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3						
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)						
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)						
		58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0								
		90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)								

Problem 8. Chemical equilibrium (15 points)

8.1	8.2	8.3	8.4	8.5	8.6	Total	Total
6	4	2	6	8	4	30	15

Iodine is an element essential for life and can be found in living organisms in small amounts. When heated under atmospheric pressure (101,3 kPa), iodine sublimates, that is, it changes the aggregation state from solid to vapor.



8.1. Calculate the sublimation temperature of iodine.

Species	$\Delta H_f^\circ (\text{kJ} \cdot \text{mol}^{-1})$	$S^\circ (\text{J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1})$
$\text{I}_2(\text{s})$	0,0	116,7
$\text{I}_2(\text{g})$	62,3	260,7

When dichloromethane (CH_2Cl_2) is added to a solution of iodine in water, iodine is distributed among the aqueous and organic phases according to the following equilibrium.

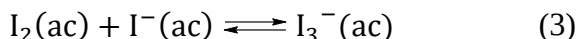


8.2. Calculate the equilibrium constant for the distribution of iodine among water and dichloromethane at 298 K. $\Delta G_f^\circ (\text{kJ} \cdot \text{mol}^{-1})$: $\text{I}_2(\text{ac}) = 16,4$; $\text{I}_2(\text{CH}_2\text{Cl}_2) = 4,00$.

Note: If you did not solve the previous item, consider the constant of distribution equal to 200 when solving the rest of the problem.

8.3. Why is iodine more soluble in organic substances than in water?

In an aqueous solution, there is an equilibrium involving the species iodine, iodide and triiodide, according to the following equation.



A suitable method to determine the solution concentrations of I_2 and I_3^- is to titrate with a standard solution of thiosulfate ($\text{S}_2\text{O}_3^{2-}$). When these iodine species interact with thiosulfate, an oxidation-reduction reaction occurs, forming iodide and tetrathionate ($\text{S}_4\text{O}_6^{2-}$).

8.4. Write the equations for the chemical reactions that happen during the titration of I_2 and I_3^- with thiosulfate. Which is the oxidation number of sulfur in thiosulphate ion?

Aiming to determine the constant for equilibrium (3), the following experiments were performed at 298 K.

- 50,0 mL of an aqueous solution of KI $0,010 \text{ mol} \cdot \text{L}^{-1}$ were added to 25 mL of I_2 solution in CH_2Cl_2 , forming two separate phases, one aqueous and the other organic.
- Through a spectrophotometric method, the iodine concentration in the organic phase was determined to be $1,20 \cdot 10^{-2} \text{ mol} \cdot \text{L}^{-1}$.
- An aliquot of 25,0 mL from the aqueous phase was taken and mixed with an excess of KI solution to avoid the evaporation of iodine. Then, the resultant solution was titrated with $\text{Na}_2\text{S}_2\text{O}_3$ $0,0100 \text{ mol} \cdot \text{L}^{-1}$ solution. The equivalence point was reached after adding 3,10 mL of the titrant.

8.5. Calculate the equilibrium concentrations of I^- , I_2 y I_3^- in the aqueous phase and the equilibrium constant for the formation of triiodide ion (3) at 298 K.

It is possible to calculate the constant for equilibrium (3) from electrochemical data. Having the following standard potential electrodes.

$E^\circ(\text{V})$: $\text{I}_3^-/\text{I}^- = 0,536$; $\text{I}_2/\text{I}^- = 0,621$.

8.6. Calculate the equilibrium constant for the formation of triiodide ion (3) at 298 K employing the data of standard electrode potential.

Problem 9. Titrations (15 points)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	Total	Total
5	5	2	2	2	3	2	9	30	15

Oxalic acid is a substance widely used as a primary standard in acid-base titrations. It is an acid of moderate strength with dissociation constants in water at 25 °C, equal to $K_{a1} = 1,41 \cdot 10^{-2}$ and $K_{a2} = 1,55 \cdot 10^{-4}$.

9.1. Calculate the pH of a solution of $\text{H}_2\text{C}_2\text{O}_4$ 0,0100 mol·L⁻¹.

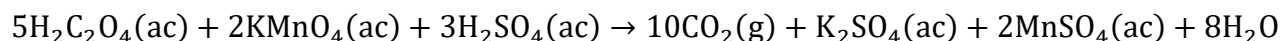
9.2. Calculate the pH of a KHC_2O_4 0,0100 mol·L⁻¹ solution.

One of the experimental challenges of the participant in the X Central American and VIII Caribbean Chemistry Olympiad (Havana, July 17th-23rd, 2016) consisted in the analysis of a mixture of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ and $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$ by a combination of acid-base and redox titrations. In the following, the experiments that could have been conducted by one of the instructors during the preparation for the practical exam are described.

A mixture of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ ($M = 126,08 \text{ g} \cdot \text{mol}^{-1}$) and $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$ ($M = 184,43 \text{ g} \cdot \text{mol}^{-1}$) was rinsed into a volumetric flask with distilled water and was completed up to the top to prepare exactly a solution containing 250,0 mL (problem solution).

25,0 mL of the problem solution were taken and transferred into a 100 mL Erlenmeyer flask. A few drops of phenolphthalein ($\text{p}K_a = 9,4$) were added and the mixture was titrated with a solution of 0,100 mol·L⁻¹ sodium hydroxide from a Mohr burette. The volume consumed from the titrant was 19,0 mL on average.

5,00 mL of the problem solution were taken and transferred to a 100 mL Erlenmeyer flask. 10 mL of 2,6 mol·L⁻¹ sulfuric acid were added, the mixture was carefully heated and titrated with a 0,00990 mol·L⁻¹ potassium permanganate solution from an amber burette. The volume consumed of the titrant was 11,2 mL on average. The reaction that occurs in this case is:



9.3. Write the chemical equation corresponding to the titration with sodium hydroxide.

9.4. The solutions of sodium hydroxide and potassium hydroxide employed during the titration were standardized previously to carry out the experiments because they are not primary standards. Mention four characteristics that must have a substance to be considered a primary standard.

9.5. How can you detect the end point of each titration?

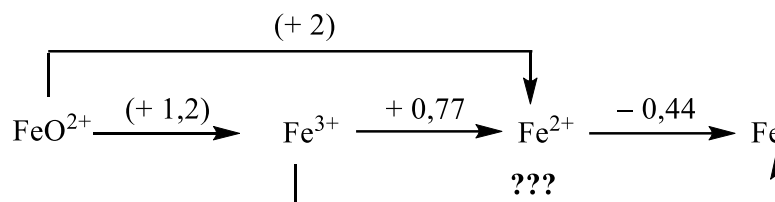
- 9.6. Demonstrate that in the conditions where the redox titration is performed, practically all the oxalate of the sample is in the form of oxalic acid. Consider that the pH value is 0,0.
- 9.7. Write the half-reactions of oxidation and reduction corresponding to the redox process.
- 9.8. Calculate the composition (in mass percent) of the mixture analyzed by the instructor.

Problem 10. Electrochemistry (20 points)

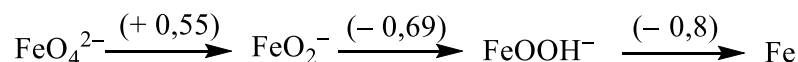
10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	Total	Total
4	4	4	4	6	6	6	6	40	20

The following Latimer diagram corresponds to iron species in acid and basic media.

Acid media



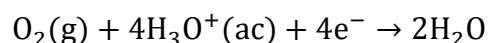
Basic media



- 10.1. In which media is Fe(III) more oxidant? In which media metallic Fe is more reductant?
- 10.2. Does any of the iron species disproportionate? Justify.
- 10.3. May species Fe(III) and Fe coexist simultaneously in an acid solution? Justify numerically.
- 10.4. Calculate the standard potential $E^\circ(\text{Fe}^{3+}/\text{Fe})$.

It is possible to understand the redox processes occurring in natural waters with the aid of electrochemical data. For example, in a lake, cold in the depth and warmer on the surface, the temperature gradients avoid vertical mixing. On the surface water is completely oxygenated and iron is present in the form of insoluble particles of $\text{Fe}(\text{OH})_3$. These particles tend to sink. More profound, the content of O_2 is low. If the organic content and other sources of reducing agent are high enough, the hydroxide will be reduced, and the iron will dissolve in the form of Fe^{2+} . Then, the formed ions will diffuse to the surface, where they will be in contact with O_2 and oxidize again to $\text{Fe}(\text{OH})_3$.

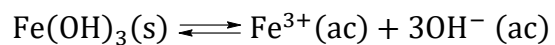
10.5. Estimate the approximate potential on the surface of an ordinary lake. Consider that the temperature is 25 °C, the pressure is 1,01 bar and the pH is 7,0. Air composition is N_2 78%, O_2 21% and Ar 1% (volume percent).



$$E^\circ = 1,23 \text{ V}$$

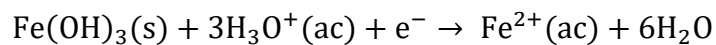
In fact, natural water conditions are slightly different from those calculated by you due to the presence of several species in the solution.

10.6. Calculate Fe^{3+} and Fe^{2+} concentrations on the surface of the lake. Consider that the temperature is 25 °C, the pH is 6,5 and the potential is 0,65 V.



$$K_{\text{ps}} = 2,79 \cdot 10^{-39}$$

10.7. Demonstrate that the standard potential corresponding to the following solution process that occurs close to the bottom of the lake is $E^0 = 0,97 \text{ V}$.



10.8. Calculate Fe^{3+} and Fe^{2+} concentrations at the bottom of the lake. Consider that the temperature is 15°C , the pH is 6,0 and the potential is 0,15 V. Overlook the effects of the temperature on the equilibrium constant and the standard potential.